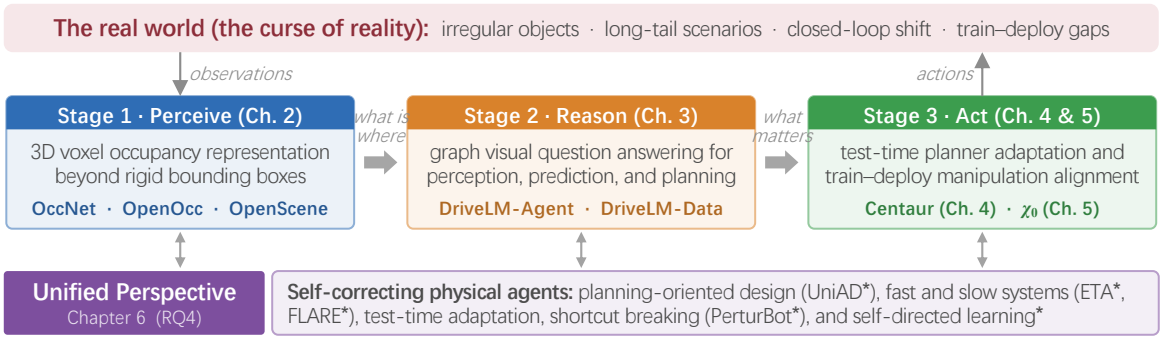

The real world (the curse of reality): irregular objects · long-tail scenarios · closed-loop shift · train–deploy gaps
observations
actions
Stage 1 · Perceive (Ch. 2)
3D voxel occupancy representation beyond rigid bounding boxes
OccNet · OpenOcc · OpenScene
what is where
Stage 2 · Reason (Ch. 3)
graph visual question answering for perception, prediction, and planning
DriveLM-Agent · DriveLM-Data
what matters
Stage 3 · Act (Ch. 4 & 5)
test-time planner adaptation and train–deploy manipulation alignment
Centaur (Ch. 4) · χ_0 (Ch. 5)
Unified Perspective
Chapter 6 (RQ4)
Self-correcting physical agents: planning-oriented design (UniAD*), fast and slow systems (ETA*, FLARE*), test-time adaptation, shortcut breaking (PerturBot*), and self-directed learning*